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Wing geometric morphometrics for identification of mosquito species (Diptera: Culicidae) of neglected epidemiological importance

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ABSTRACT

Culicidae comprises more than 3500 species, some of which are responsible for the spread of various human diseases, causing millions of deaths worldwide. Correct identification of these species is essential for the development of surveillance and control strategies. The most common method of mosquito identification is based on specific traits of the external morphology of species. However, identification of mosquitoes by morphological characters can be inaccurate or even unfeasible if the specimen is damaged or there is a lack of distinguishing features, as in the case of cryptic species complexes. Wing geometric morphometrics is a reliable, affordable tool for the identification of mosquito species, including sibling species. More importantly, it can be used in addition to both traditional morphologic identification methods as well as genetic approaches. Here, wing geometric morphometrics was used to identify sixteen mosquito species from eight genera: *Aedes, Coquillettidia, Culex, Limatus, Mansonia, Psorophora, Runchomyia*, and *Wyeomyia*. The 390 specimens used here were collected in São Paulo, Brazil using CDC traps, aspiration, and Shannon traps. Allometry was assessed by multivariate regression of the Procrustes coordinates on centroid size followed by canonical variate analysis and a pairwise cross-validated reclassification test. A Neighbor-Joining tree based on Mahalanobis distances was constructed with 1,000 bootstrap replicates using MorphoJ 1.02 and Past 2.17c. The canonical variate analysis of genera resulted in distinct clusters for *Culex, Limatus,* and *Psorophora* and partial overlapping between *Aedes, Coquilettidia*, and *Mansonia*, and between *Runchomyia* and *Wyeomyia*. Pairwise cross-validated reclassification tests indicated that genera were identified with an accuracy of at least 99% and subgenera with a mean accuracy of 96% and that in 160 of the 240 possible comparisons species were identified with an accuracy of 100%. Our results show that the eight genera in the study were correctly distinguished by wing shape, as were subgenera and most species, demonstrating that wing geometric morphometrics can be used for the identification of the mosquito species studied here.

1. Introduction

Mosquitoes (Diptera: Culicidae) belong to a large group of insects, with more than 3,500 described and validated species (Harbach, 2018; Orlandin et al., 2017). Some mosquito species are obligatory blood feeders and play a significant role in the transmission of vector-borne diseases (VBDs) (Harbach, 2018; WHO, 2017a). Malaria, dengue, Zika, yellow fever, and chikungunya endanger individuals living in endemic areas, i.e., roughly 80% of the world's population. The majority of endemic areas are in developing countries and poor regions of the world, where pathogens transmitted by mosquitoes cause millions of cases of these diseases every year (Benelli and Beier, 2017; Etienne et al., 2017; Faria et al., 2017; Gubler, 2011; Paules and Fauci, 2017; Sinka et al., 2012; Wilke et al., 2019; World Health Organization - WHO, 2017b).

The presence of many arboviruses has been documented in Brazil. Dengue, Zika, yellow fever, and chikungunya viruses are the most commonly found and have the greatest epidemiological relevance. However, many other neglected arboviruses are circulating under the radar such as Mayaro, Oropouche, and Rocio viruses (Lima-Camara, 2016; Lorenz et al., 2017b; Mourão et al., 2009).

Major outbreaks caused by different arboviruses have been reported in Brazil in the recent past. Between 2016 and 2018 a major yellow fever outbreak resulted in approximately 2000 human cases. Even though *Ae. aegypti* is acknowledged as the main yellow fever vector in

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urban areas, *Aedes serratus* is considered a secondary vector, and its importance in the maintenance of the outbreak is still poorly understood (Giovanetti et al., 2019; Paules and Fauci, 2017).

Aedes aegypti and *Ae. albopictus* have been implicated in major recent outbreaks of Zika, chikungunya, and dengue viruses. Zika virus was reported for the first time in Brazil in 2015 causing approximately 150,000 cases (Faria et al., 2016; Lima-Camara, 2016; Zanluca et al., 2015). The chikungunya virus was first detected in Brazil in 2014 in the state of Amapa, and since then has spread throughout Brazil (Lima-Camara, 2016; Nunes et al., 2015). Since the first major dengue outbreak in Brazil in 1986, Brazil has been endemic for dengue with all four serotypes of the virus circulating in the country. In 2019 alone, 1.2 million cases of dengue were reported in Brazil (Nunes et al., 2019; Pan American Health Organization/World Health Organization, 2019; Benelli et al., 2020; Achee et al., 2019)

Even though the most epidemiologically important arboviruses circulating in Brazil are transmitted by *Ae. aegypti* and *Ae. albopictus*, yellow fever, dengue, and Zika viruses have been isolated in *Ae. serratus, Aedes terrens*, subgenus *Melanoconion* of *Culex, Limatus durhamii, Mansonia indubitans, Mansonia titillans, Psorophora ferox*, and *Wyeomyia confusa* (Barrio-Nuevo, 2019; Cardoso et al., 2010a, b; Figueiredo, 2007; Lourenço-de-Oliveira and Failloux, 2017; Méndez et al., 2001; Mourão et al., 2009; Santos-Neto and Lozovei, 2008; Segura et al., 2007; Turell et al., 2000; Unlu et al., 2010; Vasconcelos et al., 1998). Furthermore, reliable and up to date information on the vectorial capacity of many of the mosquito species in Brazil is lacking as highlighted by the recent finding of the susceptibility of *Aedes fluviatilis* to the chikungunya virus (de Melo Ximenes et al., 2020).

It is widely accepted that the most effective way to prevent and control VBD outbreaks is by controlling the population of vector mosquitoes (Fernandes et al., 2018; Wilke et al., 2018). As different species of mosquitoes often have distinct ecologies and behaviors and therefore play different roles in the transmission of VBDs, the correct identification of vector species is crucial to establish the epidemiological profile of disease transmission in a given area (Farajollahi et al., 2011; Githeko et al., 1996).

The most common method for mosquito identification is based on specific external morphological traits (Darsie and Morris, 2000). However, the identification of specimens based on morphological traits may not be possible if specimens are damaged or lack scales or bristles (Wilke et al., 2016). Furthermore, it can be extremely difficult to distinguish between very similar species and between members of cryptic species complexes (e.g., the *Culex pipiens* and *Anopheles gambiae* complexes) as species frequently differ only in the male genitalia, making correct identification of female specimens particularly challenging (Börstler et al., 2014; Calado and Navarro-Silva, 2005; Farajollahi et al., 2011; Fonseca et al., 2004; Lorenz et al., 2012; Lorenz et al. 2017a; Petrarca et al., 1998; White, 1974).

Molecular markers can be useful in cases where the morphological identification is compromised and has already been successfully used to differentiate *Anopheles* (*Kerteszia*) females, representants of the Culicidae subfamily, larvae and eggs of *Aedes* mosquitoes and to differentiate *Culex* mosquitoes (Araki et al., 2016; Beebe et al., 2007; Calado and Navarro-Silva, 2005; Engdahl et al., 2014; Kirchgatter et al., 2020; Lorenz et al., 2015; Toma et al., 2000; Weeraratne et al., 2018). However, molecular markers can be relatively costly and need specific infrastructures (Dujardin, 2008).

Wing geometric morphometrics (WGM) is a cost-effective technique widely used in population genetics and phylogeny studies (Christe et al., 2019; de Carvalho et al., 2017; Lorenz et al., 2017a, 2014, 2012; Multini et al., 2019; Vidal et al., 2011; Wilk-da-Silva et al., 2018) and has also been proven reliable for mosquito identification (Jaramillo-O et al., 2015; Laurito et al., 2015; Wilke et al., 2016; Sontigun et al., 2017). WGM can be used with traditional morphological keys to identify mosquito species and is a great asset when identification by taxonomic keys is made difficult or even unfeasible because of the inherent limitations of the technique or absence of specific morphological structures (Calle et al., 2002; Vidal et al., 2011; Lorenz et al., 2012; Wilke et al., 2016). WGM can aid mosquito identification and can help accelerate the species identification process, especially in the case of species for which morphological identification is problematic. In light of this, we hypothesized that wing shape variation can be used to correctly identify mosquito species of neglected epidemiological importance in Brazil. The objective of this study was, therefore, to use WGM to identify sixteen mosquito species from eight different genera.

2. Material and methods

2.1. Mosquito sampling and identification

Adult specimens were collected in the city of São Paulo, Brazil, in two conservation areas, Capivari-Monos Environmental Preservation Area (Capivari-Monos EPA) (23° 54.556′S 46° 42.167′W) and Cantareira State Park (23° 26.865′S 46° 38.091′W). Mosquitoes were monthly collected from March 2015 to April 2016. Mosquitoes were collected using CO_2-baited CDC light traps (Gomes et al., 1985), aspiration, and Shannon traps and taken to the Entomology Laboratory at the School of Public Health, University of São Paulo, where they were identified to species level with the aid of taxonomic keys (Darsie and Morris, 2000). After identification, specimens were stored in 1.5 mL microcentrifuge tubes with silica gel until their wings were removed (Table 1). All specimens used were females, except for six male specimens which have been added to increase sampling sufficient in rare species as follows: *Coquilletidia albifera* (2 males), *Culex pereyrai* (3 males), and *Ma. indubitans* (1 male).

The species were chosen according to the place of collection and the habitats described in the literature for each one, considering that the purpose of the study was to examine species in peridomestic environments, of known wild habits.

2.2. Data acquisition

The right wing of each specimen was detached from the thorax and mounted between a slide and coverslip with Canada balsam (Sigma-Aldrich, St. Louis, MO, USA). The wings were then photographed with a Leica DFC 320 digital camera under 40x magnification, in a stereomicroscope and 18 landmarks were digitized with tpsDig V2.32 (Rohlf, 2015) (Fig. 1).

2.3. Geometric morphometrics analysis

To assess allometry (the effect of wing size on wing shape) we performed multivariate regression of the Procrustes coordinates on CS using a permutation test with 10,000 randomizations. A discriminant analysis was performed to determine the degree of dissimilarity between species in a morphospace defined by canonical variate analysis (CVA) and to calculate the Mahalanobis distances

Each specimen was then reclassified according to the similarity of its wings to the average shape for all the species using a pairwise cross-validated reclassification test based on Mahalanobis distances to evaluate the accuracy of morphometric classification. The same analyses were carried out without the male specimens to observe possible inconsistencies with the results. All the analyses were carried out in MorphoJ 1.02. (Klingenberg, 2011). To visualize the similarities between the species, a Neighbor-Joining (NJ) tree based on the Mahalanobis distances between species was constructed using PAST 1.89 (Hammer et al., 2001) with 1,000 bootstrap replicates; 30 specimens of *Anopheles cruzii* were used as an outgroup. These specimens were collected in the city of São Paulo, Brazil, in the Capivari-Monos Environmental Preservation Area, in an urban environment.

Table 1
Information on mosquito species collected, including collection year.

Taxon	N	Collection Year	Epidemiological Importance	Morphological Identification (Harbach, 2018)	Coordinates
Aedes (Ochlerotatus) crinifer (Theobald, 1903)	30	2015/2016	Unknown	Morphological characters of adults; male and female genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Aedes (Ochlerotatus) serratus (Theobald, 1903)	30	2015/2016	Encephalitis*; Yellow Fever; Oropouche (Vasconcelos et al., 1998; Figueiredo, 2007; Segura and Castro, 2007; Mourão et al., 2009; Cardoso et al., 2010a,b; Laporta et al., 2012)	Morphological characters of adults; male and female genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Aedes (Protomacleaya) terrens (Walker, 1856)	30	2015/2016	Chikungunya (Lourenço-de-Oliveira and Failloux, 2017)	Morphological characters of adults; male genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Coquillettidia (Rhynchotaenia) albifera (Prado, 1931)	21	2015/2016	Unknown	Morphological characters of adults; male genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Culex (Melanoconion) pereyrai (Duret, 1967)	11	2015	Unknown	Morphological characters of adults; male genitalia.	23° 26.865′S 46° 38.091′W
Culex (Melanoconion) ribeirensis (Forattini & Sallum, 1985)	30	2015/2016	Encephalitis* (Santos-Neto and Lozovei, 2008)	Morphological characters of adults; male genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Culex (Melanoconion) vaxus (Dyar, 1920)	22	2015/2016	Dengue (Barrio-Nuevo, 2019)	Morphological characters of adults; male genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Limatus durhamii (Theobald, 1901)	26	2015/2016	Zika (Barrio-Nuevo, 2019)	Morphological characters of adults.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Mansonia (Mansonia) indubitans (Dyar & Shannon, 1925)	30	2015/2016	Encephalitis* (Turell et al., 2000)	Morphological characters of adults; male and female genitalia.	23° 54.556′S 46° 42.167′W
Mansonia (Mansonia) titillans (Walker, 1848)	10	2015	Encephalitis* (Méndez et al., 2001; Unlu et al., 2010)	Morphological characters of adults; male and female genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Psorophora (Janthinosoma) ferox (Humbolt, 1819)	30	2015/2016	Encephalitis*, Mayaro, Yellow Fever (De Lopes et al., 1981; Mitchell et al., 1986; Bernard et al., 2001; Moreno et al., 2011)	Morphological characters of adults; male genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W
Runchomyia (Runchomyia) reversa (Lane & Cerqueira, 1942)	30	2015/2016	Unknown	Morphological characters of adults; male genitalia	23° 54.556′S 46° 42.167′W
Wyeomyia (Prosopolepis) confusa (Lutz, 1905)	30	2015/2016	Zika virus (Barrio-Nuevo, 2019)	Morphological characters of adults; male genitalia.	23° 26.865′S 46° 38.091′W
Wyeomyia (Phoniomyia) edwardsi (Lane & Cerqueira, 1942)	21	2015	Unknown	Morphological characters of adults; male genitalia.	23° 54.556′S 46° 42.167′W
Wyeomyia (Miamyia) oblita (Lutz, 1904)	20	2015/2016	Unknown	Morphological characters of adults; male genitalia.	23° 54.556′S 46° 42.167′W 23° 26.865′S 46° 38.091′W

* Eastern equine encephalitis; West Nile encephalitis; Rocio viral encephalitis; Saint-Louis encephalitis; Venezuelan equine encephalitis.

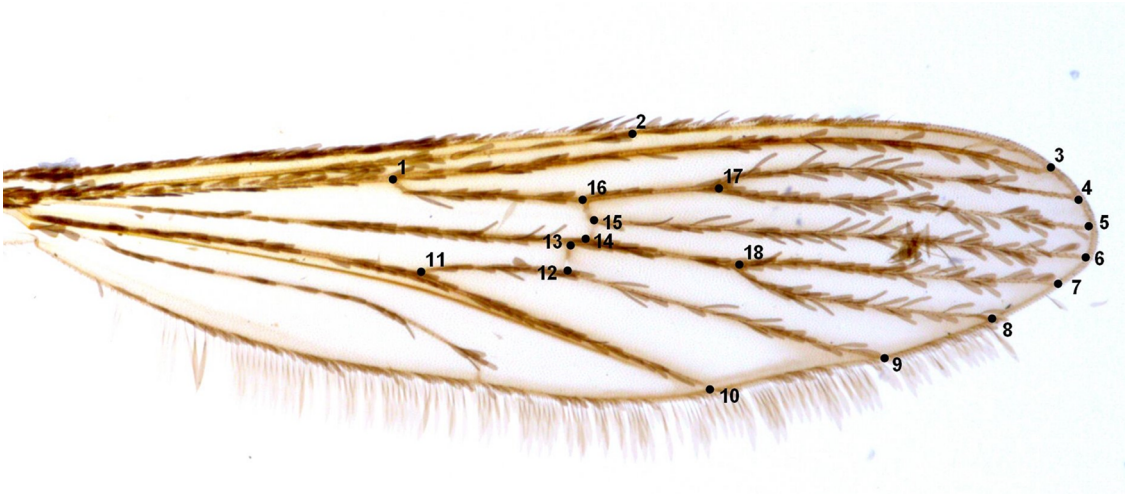

Fig. 1. Right wing of *Wyeomyia edwarsi* showing the 18 landmarks used in this study.

3. Results

The allometric effect of wing size on wing shape variation was 4.26% ($P < 0.0001$); although small, this was not excluded from the analysis because we considered it informative and part of the species-identification process (Dujardin, 2008; Wilke et al., 2016).

Canonical variate analysis (CVA) of wing shape variation of the sixteen species showed a degree of overlap between some species, belonging to the same genus and similar to each other. The analysis generated distinct clusters for the species *Li. durhamii* and *Ps. ferox* (Supplementary Figure 1). The CVA of wing shape for each of the eight genera resulted in distinct clusters for *Culex, Limatus*, and *Psorophora* in the morphospace and partial overlap between *Aedes, Coquillettidia,* and *Mansonia* and between *Runchomyia* and *Wyeomyia* (Fig. 2).

CVA was then performed on the data for each species of each genus individually, confirming the identification of subgenera (when present) and species. Pair-wise wireframe graphs for all species are displayed next to their respective CVA.

The *Aedes* species grouped into three clusters, correctly identifying the subgenera *Ochlerotatus* and *Protomacleaya* as well as the three species analyzed, *Aedes crinifer, Ae. serratus* and *Ae. terrens*. A minor degree of overlap was observed between *Ae. serratus* and *Ae. crinifer,* caused by only one specimen, which is consistent with the fact that they are from the same subgenus, *Ochlerotatus* (Fig. 3A). The genus *Wyeomyia* was segregated into three main clusters, one comprising *Wy. confusa*, another *Wyeomyia edwardsi,* and the third *Wyeomyia oblita* and *Wyeomyia serratoria* (for the latter of which the subgenus has yet to be elucidated (Gaffigan et al., 2001) with partial overlap. The three subgenera, *Prosopolepis, Phoniomyia*, and *Myamyia*, were correctly distinguished (Fig. 3B). The *Culex* genus grouped into three distinct clusters that correctly segregated the three *Culex* species analyzed in this study, *Culex ribeirensis, Cx. pereyrai* and *Culex vaxus* (Fig. 3C). Both species from the genus *Mansonia* clustered correctly in the morphospace with minimum overlap between them. Furthermore, a comparison of *Mansonia* species *Ma. indubitans* and *Ma. titillans* with *Cq. albifera* also resulted in their correct clustering and subsequent identification in the CVA morphospace (Fig. 3D).

The NJ tree yielded similar results to those observed for the CVA, and the species grouped correctly according to genera and subgenera (Fig. 4). *Limatus durhamii* was the most distinct species. All the species from the genus *Culex* grouped together, with higher levels of similarity between *Cx. ribeirensis* and *Cx. vaxus*. The same pattern was observed for the genus *Aedes*: all the species grouped together, but *Ae. crinifer* and *Ae. serratus* had greater similarity with each other than with the remaining species (both species belong to the subgenus *Ochlerotatus*). The *Mansonia* species also grouped together and showed some similarity with *Cq. albifera*. These three species belong to the tribe Mansoniini. The *Wyeomyia* species clustered together and divided between two branches, one comprising *Wy. edwardsi* and *Wy. confusa* and the other comprising *Wy. oblita* and *Wy. serratoria*.

The pairwise cross-validated reclassification test resulted in high values for most species. When identification was carried out at the genus level, the test resulted in values no lower than 90% and yielded a perfect reclassification (100%) in 48 out of 56 comparisons (Table S1). When the analyses were carried out without the male specimens, no further differences were observed (Table S2).

At the species level, in 160 out of a total of 240 comparisons (120 comparisons below the diagonal, corresponding to group 1 compared with group 2 and 120 comparisons above the diagonal, corresponding to group 2 compared to group 1), the species were correctly identified, corresponding to a reclassification score of 100%. The average reclassification score was 96%, indicating that WGM is a highly reliable technique for the identification of these species. Only three reclassification scores were below 70%: *Ma. indubitans* vs. *Ma. titillans* (63%), *Ma. titillans* vs. *Ma. indubitans* (50%) and *Wy. serratoria* vs. *Wy. oblita* (65%) (Table 2). However, despite these isolated low reclassification scores, *Ma. indubitans, Ma. titillans* and *Wy. serratoria* had mean reclassification scores of 92%, 97%, and 95%, respectively.

4. Discussion

WGM has been proven to be a reliable, efficient tool for the identification of mosquitoes (Changbunjong et al., 2016; Jaramillo-O et al., 2015; Sontigun et al., 2017; Wilke et al., 2016). Our findings reflect the effectiveness of WGM in correctly identifying mosquito species solely by their wing shape variation. The technique was powerful enough to identify correctly genera, subgenera, and most species with high accuracy, allowing us to identify with a high degree of confidence eight genera comprising sixteen neglected mosquito species, some of these with described epidemiological importance: *Ae. crinifer, Ae. serratus, Ae. terrens, Cq. albifera, Cx. pereyrai, Cx. ribeirensis, Cx. vaxus, Li. durhamii, Ma. indubitans, Ma. titillans, Ps. ferox, Runchomyia reversa, Wy. confusa, Wy. edwardsi, Wy. oblita* and *Wy. serratoria*.

In our analyses, the mosquito species grouped correctly according to their genera and subgenera. CVA showed some degree of overlap between the genera *Mansonia, Coquillettidia* and *Aedes*, with the first two belonging to the tribe Mansoniini and the last to the tribe Aedini; these tribes are known to share similarities with each other (Harbach, 2018; Harbach and Reinert, 1998; Reinert et al., 2004). A further overlap can be seen between *Runchomyia* and *Wyeomyia,* both of which belong to

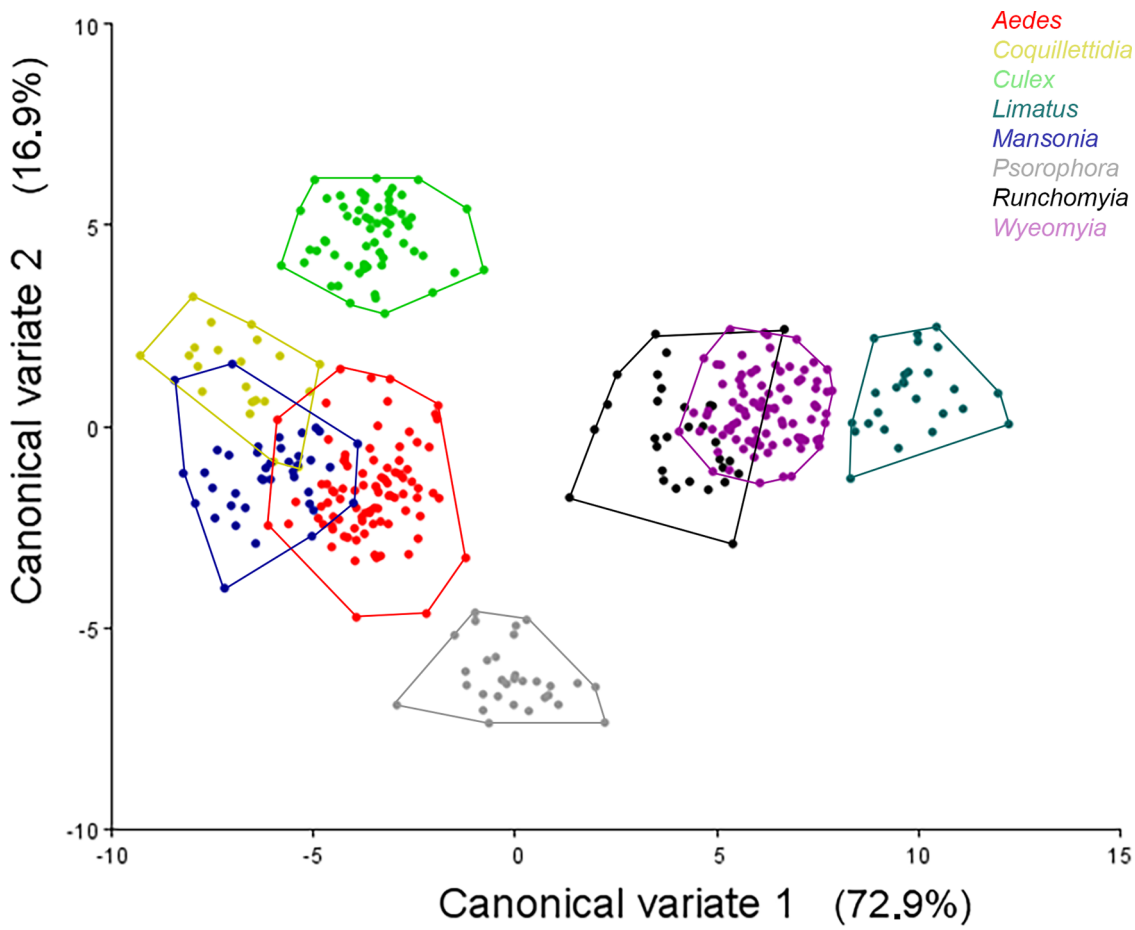

Fig. 2. Morphological space produced by CVA of wing shape variation of the eight mosquito genera.

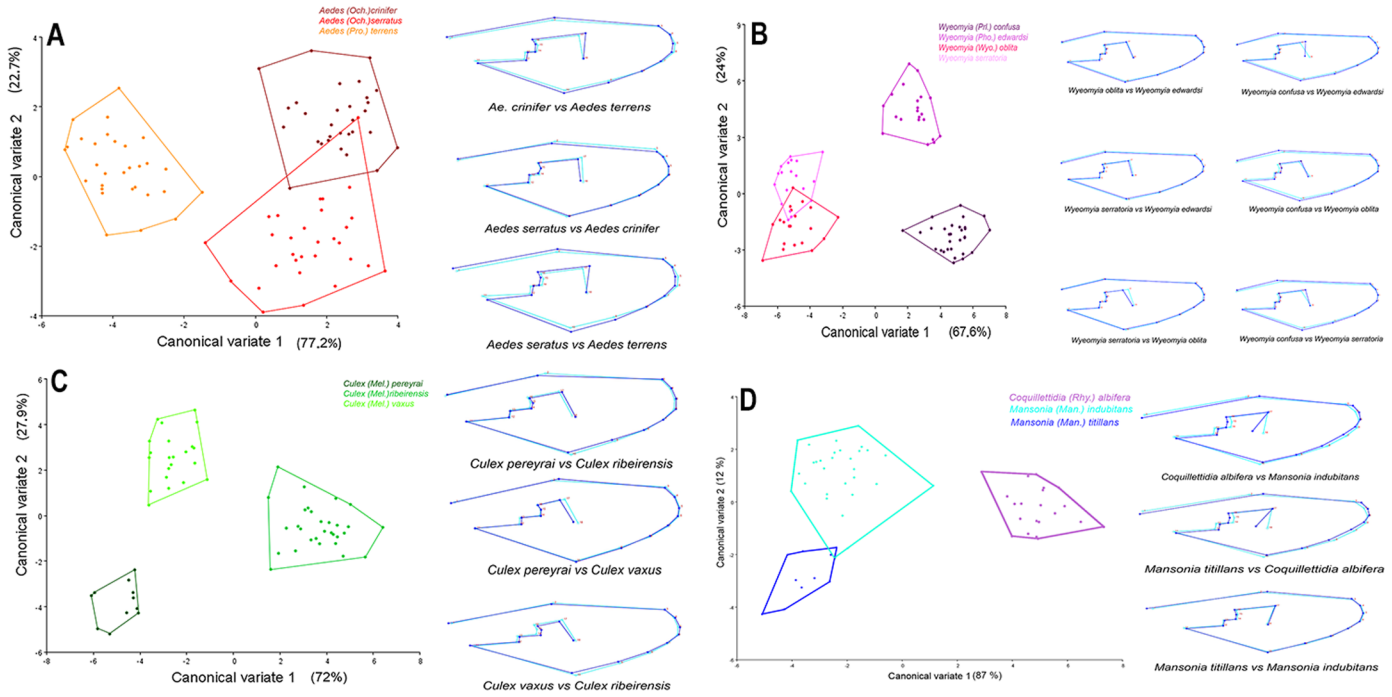

Fig. 3. Comparison of wing shapes in the morphospace based on Mahalanobis distances and shape variations in a wireframe graph. (A) *Aedes* species; (B) *Wyeomyia* species; (C) *Culex* species; and (D) *Mansonia* and *Coquillettidia* species.

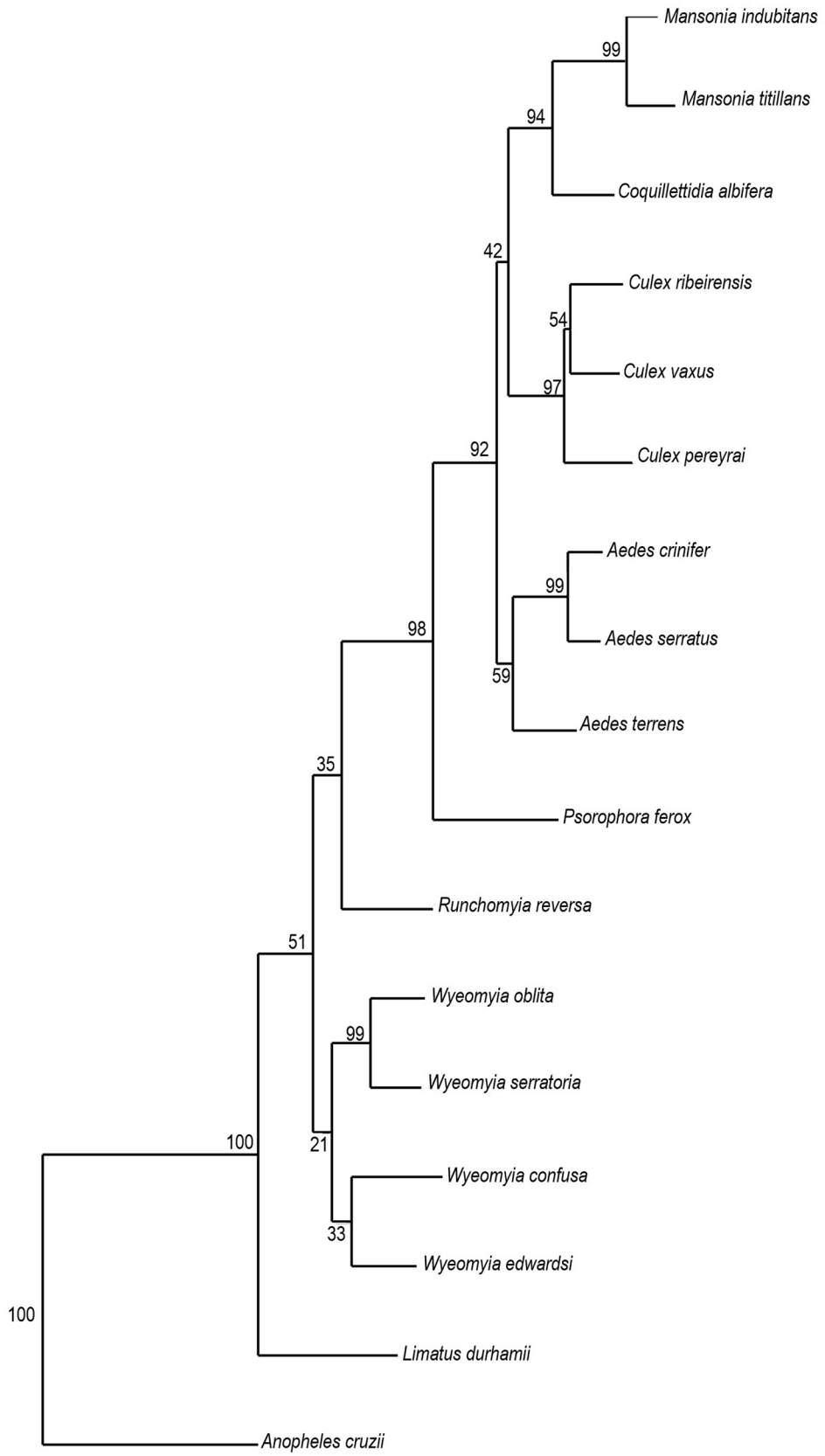

Fig. 4. Neighbor-Joining tree based on Mahalanobis distances with 1000 bootstrap replicates.

the tribe Sabethini. The species from the genera *Culex, Limatus,* and *Psorophora* exhibited greater dissimilarity, forming distinct groups in the CVA morphospace. These results were corroborated by the NJ tree.

The difference in the values on the diagonals observed in the pairwise cross-validated reclassification test is due to natural variations in the wing patterns in the species compared. When species share similar wing shape patters but one of the species has higher levels of wing shape variability and the other species has lower levels of wing shape

Table 2
Results of pairwise cross-validated reclassification testes.Group 2.

		Ae. crinifer	*Ae. serratus*	*Ae. terrens*	*Cq. albifera*	*Cx. pereyrai*	Cx. ribeirensis	*Cx. vaxus*
Group 1	***Ae. crinifer***	x	73	97	100	100	97	100
	Ae. serratus	77	x	90	100	91	100	100
	Ae. terrens	97	93	x	90	91	97	100
	Cq. albifera	100	100	100	x	100	100	95
	Cx. pereyrai	97	86	83	100	x	80	73
	Cx. ribeirensis	100	100	100	100	82	x	91
	Cx. vaxus	97	100	97	100	73	80	x
	Li. durhamii	100	100	86	100	100	100	100
	Ma. indubitans	93	96	100	86	82	96	100
	Ma. titillans	93	86	90	95	100	100	82
	Ps. ferox	100	96	100	100	100	100	100
	Ru. reversa	100	100	100	100	91	100	100
	Wy. confusa	100	100	100	100	100	100	100
	Wy. edwardsi	100	100	100	100	91	100	100
	Wy. oblita	100	100	100	100	100	100	100
	Wy. serratoria	100	100	100	100	100	100	95

	Li. durhamii	*Ma. indubitans*	*Ma. titillans*	*Ps. ferox*	*Ru. reversa*	*Wy. confusa*	*Wy. edwardsi*	*Wy. oblita*	*Wy. serratoria*
Group 1	100	90	100	100	100	100	100	95	100
	100	87	90	100	100	100	100	100	100
	100	97	70	100	100	100	100	95	100
	100	80	80	100	100	100	91	95	95
	100	97	100	100	90	100	100	100	95
	100	93	100	100	100	100	95	100	100
	100	100	70	100	100	100	100	100	100
	x	100	100	100	100	97	100	85	100
	100	x	50	100	100	100	100	95	100
	85	63	x	97	100	100	100	100	100
	100	93	90	x	100	100	100	100	100
	100	100	100	100	x	100	90	85	100
	100	100	100	100	97	x	86	95	100
	100	100	100	100	90	87	x	90	89
	92	90	90	100	100	100	100	x	79
	96	100	100	100	100	100	100	65	x

$P < 0.001$. Values are given in percentages.

variability, the first will yield higher values in the pairwise cross-validated reclassification test, whereas the latter will yield lower values.

When the pairwise cross-validated reclassification test was carried out without the six male specimens no further differences were observed in the results and the reclassification scores remained high in all the comparisons, however the reclassification score in the comparison between *Cx. pereyrai* and *Ae. crinifer* decreased considerably (from 100% to 75%), but the score is considered high.

The analysis within each genus was also able to identify species correctly. The species from the genus *Aedes* were correctly identified with a high degree of confidence, with minor overlapping between *Ae. serratus* and *Ae. crinifer*. These two species belong to the subgenus *Ochlerotatus,* which may explain the overlap between them (Gaffigan et al., 2001). We believe that the overlapping between these two species is due to wing shape variation that is naturally found in species from the same subgenus. *Aedes serratus* is difficult to identify by morphological characters and used to be considered a synonym of *Ae. nubilus* (Lane, 1992). A species of epidemiological importance, *Ae. serratus* been found naturally infected by Ilheus, Saint-Louis, Oropouche and yellow fever viruses (Cardoso et al., 2010a, b; Figueiredo, 2007; Laporta et al., 2012; Mourão et al., 2009; Segura et al., 2007; Vasconcelos et al., 1998, 1991). As there are no records of these viruses naturally infecting *Ae. nubilus*, correct identification of *Ae. serratus* is important to determine their transmission cycle.

Culex species were also correctly identified in CVA morphospace even though they all belong to the same subgenus, *Melanoconion*. This subgenus poses many obstacles to traditional morphological identification because of the high levels of similarity between species, and in many cases, the male genitalia is the only morphological character that can be used for reliable identification of species within the subgenus (Pecor et al., 1992; Torres-Gutierrez and Sallum, 2015). *Melanoconion* comprises two sections: the Spissipes section, possibly of greater epidemiological importance, and the Melanoconion section. The latter has a more complex morphology and is, therefore, more difficult to identify to species level (Sallum and Forattini, 1996; Harbach, 2018). Here, species from both sections were analyzed (*Cx. pereyrai* and *Cx. ribeirensis*, from Spissipes, and *Cx. vaxus*, from Melanoconion) and were correctly identified by WGM. The reclassification scores were higher than 70% in all the possible comparisons, and the CVA analysis showed no overlap between the three species, proving that WGM is a reliable tool for identification of the species from genus *Culex* used in this study.

Although WGM has been successfully used in other studies that focused on the identification of species from the genus *Culex* (Börstler et al., 2014; Laurito et al., 2015), this is the first study to use this technique to identify species within the subgenus *Melanoconion*. Some species of this subgenus are of epidemiological importance as they have been found infected by arboviruses. The species *Cx. ribeirensis*, which is involved in the transmission of Eastern equine encephalitis (EEE), was originally incorrectly identified as *Cx. epanastasis*, which in turn was long misidentified as *Culex crybda* (Santos-Neto and Lozovei, 2008). Because of the challenge in identifying species belonging to the *Melanoconion* subgenus, when a mosquito of this subgenus is found infected with pathogens, identification is frequently only possible to subgenus level (Auguste et al., 2010; Deardorff and Weaver, 2010; Iversson, 1994; Sallum and Forattini, 1996; Turell et al., 2005), demonstrating the importance of a new tool to correctly identify mosquitoes and determine the epidemiological cycles of arboviruses.

Here, WGM also proved to be a reliable tool for the identification of *Wyeomyia* species. The CVA morphospace formed by the four species of this genus showed that they clustered by subgenus. *Wyeomyia confusa* and *Wy. edwardsi* separated into two completely distinct clusters, while *Wy. oblita* and *Wy. serratoria* showed some degree of overlap. The subgenus of *Wy. serratoria* has yet to be elucidated, and the proximity of this species to *Wy. oblita* in CVA morphospace indicates some similarities between them, as well as with other species of the tribe Sabethini (e.g., *Li. durhamii* and *Ru. reversa*) (Harbach, 2007). Although *Wyeomyia* species are of more limited epidemiological importance, some have been found infected with arboviruses, indicating that this genus may play a role in the epidemiological cycle of pathogens in nature. However, this has yet to be elucidated (Barrio-Nuevo, 2019; Chowdhary et al., 2012; Roca-Garcia, 1944; Auguste et al., 2010; Turell et al., 2005).

Mansonia titillans, Ma. indubitans and *Cq. albifera* belong to the tribe Mansoniini and share many morphological characteristics (Harbach, 2018; Harbach and Reinert, 1998). WGM was able to distinguish the three species with high degrees of confidence, and despite the relatively low scores obtained in the pairwise cross-validated reclassification test for these species on their own, the comparisons with other species yielded high scores. *Ma. indubitans* and *Ma. titillans* are usually indistinguishable unless the terminalia are examined under a microscope (Lane, 1992), and WGM analysis can partially help to ensure their correct identification. *Mansonia titillans* has been found naturally infected by the EEE virus and other arboviruses that cause febrile diseases, and *Ma. indubitans* is moderately susceptible to infection with four strains of Venezuelan equine encephalitis (Méndez et al., 2001; Turell et al., 2000; Unlu et al., 2010).

Limatus durhamii and *Ps. ferox* each clustered completely apart from all the other species in all the analyses carried out. The proximity of *Ps. ferox* to the other Aedini representatives in the CVA analysis was expected, as this species belongs to the same tribe. In the NJ tree, *Ps. ferox* and *Li. durhamii* were the most distinct species, highlighting the ability of WGM to identify both species reliably and correctly. *Limatus durhamii* was recently found naturally infected by Zika virus (Barrio-Nuevo, 2019) and other viruses not associated with major epidemics (Guama, Maguari, and Tucunduba viruses) (Segura and Castro, 2007; Turell et al., 2005). *Psorophora ferox* is known to be the primary vector of Rocio encephalitis in Brazil (De Lopes et al., 1981; Mitchell et al., 1986; Vasconcelos et al., 1998) and was found infected by West Nile, Ilheus, Una, and yellow fever viruses (Bernard et al., 2001; Moreno et al., 2011; Turell et al., 2005; Vasconcelos et al., 1998). Both species are more commonly found in a natural habitat but often leave forested areas to blood-feed on humans and other animals in peri-urban areas (Lopes et al., 1993), behavior that can lead to transmission of pathogens from the sylvatic environment to humans.

In recent observations, Lorenz and Suesdek (2020) used WGM to analyze if wing shape might be phylogenetically meaningful in Culicidae family. The authors' used species also included in this study *(Cx. vaxus, Cx. ribeirensis, Cq. albifera, Ps. ferox, Li. durhamii, Ma. titillans,* and *Wy. confusa*), and demonstrated WGM robustness and applicability. The authors concluded that other morphological characters should also be used to assess the phylogeny of Culicidae.

As pointed before, molecular characterization is successfully used for mosquito identification (Beebe et al., 2007; Calado and Navarro-Silva, 2005; Engdahl et al., 2014; Toma et al., 2000; Weeraratne et al., 2018). However, molecular techniques are expensive and require specific knowledge and equipment (Garros and Dujardin, 2013; Lorenz et al., 2017a). WGM can be used alongside other mosquito identification techniques such as molecular and morphological techniques. Furthermore, results of previous works and those reflected here demonstrate that the WGM technique is reliable and informative, besides being fast and affordable (Jaramillo-O et al., 2015; Lorenz et al., 2017a).

5. Conclusions

WGM correctly identified the genera, subgenera, and species of the mosquito species analyzed in this study, including similar species that are difficult to identify based only on external morphological traits. The high degree of confidence in identifying the distinct mosquito genera and species studied here consolidates the usefulness of WGM as a taxonomic tool. This technique can be used as a reliable, useful tool for the identification of mosquito species, especially when the specimen is

damaged and identification by morphological characters is unfeasible. Furthermore, WGM can be used alongside traditional taxonomic tools such as the ones based on specific traits of the external morphology and molecular markers. Optimizing WGM for the taxonomic identification of mosquito species should be considered in the future.

CRediT authorship contribution statement

Ana Letícia da Silva de Souza: Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft. **Laura Cristina Multini:** Methodology, Validation, Writing - review & editing. **Mauro Toledo Marrelli:** Validation, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition. **André Barretto Bruno Wilke:** Methodology, Software, Validation, Formal analysis, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.actatropica.2020.105593.

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